

Components needed:

- Arduino Uno
- Bluetooth module HC-05
- Three channel relay board
- Connecting wires

The three-channel relay board and the Bluetooth module are connected to the Arduino Uno board. The smartphone app connects to the Arduino via the smartphone app. The android app then sends instructions to switch a particular light ON or OFF via the Bluetooth module. The relay board is used to turn the particular light ON or OFF. To try implementing this voice activated switch head over to <http://electronicsforu.com/electronics-projects/hardware-diy/arduino-home-automation-system-android-app> .

Security camera using Arduino Yun

Security is a great necessity for any automated home. There is a small chance that someone might break into your home. A system in place to either catch a thief or get a clue of his identity will be of help.

This DIY uses a camera connected to an Arduino Yun board. Arduino Yun is a very powerful Arduino board with on-board Wi-Fi, LAN and a card reader.

Components needed:

- USB camera
- PIR sensor
- Arduino Yun
- Connecting wires

The DIY uses a PIR (passive infrared) sensor to detect motion. When the PIR sensor detects motion, it sends a signal. You can read more about the working of a PIR sensor at <https://learn.adafruit.com/pir-passive-infrared-proximity-motion-sensor/how-pirs-work> . The board switches on the USB camera to take pictures of the moving object which are then stored on the SD card. For the detailed DIY instruction please visit the DIY link at <https://learn.adafruit.com/wireless-security-camera-arduino-yun/introduction>

Complete home automation using OpenHAB

This DIY is a complete package that uses an open source project called OpenHAB to help in home automation. OpenHAB has apps for both Android and iOS.